

Module 6

Intro to Parametric Design — Tier 2: Intermediate

1. What is Parametric Design?

Parametric CAD builds shapes from adjustable numeric values (parameters) rather than freehand drawing — change a number, the whole model updates. This makes edits fast and precise, ideal for functional parts.

2. Beginner-Friendly Tools

Tool	Best for
Tinkercad	Absolute beginners — block-based, browser-only, free
Fusion 360	Full parametric CAD — steeper learning curve, industry-standard
Shapr3D	Tablet-friendly, intuitive sketch-based modeling

3. Core Concepts

- **Sketch** — a 2D shape you draw as the base of a 3D feature
- **Extrude** — pushing a 2D sketch into 3D depth
- **Boolean operations** — combining, subtracting, or intersecting shapes
- **Fillet/chamfer** — rounding or beveling sharp edges

4. Editing Existing Models

- Scaling a downloaded model to fit your needs
- Adding text/labels to a base model
- Combining two separate models into one printable file
- Splitting large models into printable sections

End of Module 6 — Next: Module 7, Design for Printability